

Active InferAnt Stream 015.1 ~ START/HERE: A Map & What Might Happen

ActiveInferAntStream | START/HERE: A Map & What Might Happen Next | 1:03:05

all right it's september 3rd 2025 this is active inference stream 15.1 been a little bit but we're back we're in the start repo start.active inference institute and this should be a very fun multi-level stream right off the bat this is where we're going to get through today i'm running run.sh from the top of the repo and several functionalities are shown which we're going to go through the first is the system checkup the second is an environment setup step the third is the generation of custom curriculum this is going to use llm's generative ai to write custom active inference onboarding curricula depending on a given audience or entity and then translate that into whatever language you want and then the fourth option is the repository manager where we will clone into a variety of repos so i am going to get a few of those cooking welcome this fine evening looking forward to everyone's comments and questions first in this one i'm gonna get the repository cloning going clone the available repositories and get that going in the background then i'm going to open up another run.sh instance dot slash run sh on the second one i'm going to start up a custom curriculum generator as soon as somebody types in the live chat a given entity that they'd like to see i'll add in another one as well so let's get in there we will go to three generate curricula

initializing the epistemic and pragmatic aspects of consciousness

connecting to the free energy principle okay here we go

what this interactive curriculum generator gives us is uh some options some interactive options for what kind of domain we want to do research into and what audience we want to tailor the material to what language so we'll start out with either entering an existing domain like physics or we can add a new domain

this is going to be um coffee tasting

yes it's a new domain target entities we can pick a given entity we'll go with tulsi gabbard just as the first listed entity there and then we can pick a target language and we'll go with ancients greek okay it's a new language that's fine all right coffee tasting tulsi gabbard ancient greek yes proceed with that okay while that's proceeding check in on the cloning this is going into the clones subfolder under src and that is cloning into a variety of repos some of which are explored in other streams others which i've made some major updates to over the previous few days so we're going to look into some in a little bit more detail let's pull back to this agenda so again welcome this is the start slash here a map and what might happen next there's a repo that goes along with this just expanded the browser window this repo is

a github dot com slash active inference institute slash start and

it is an advanced ai powered system for creating personalized active inference and free energy principle curricula and more and much more it's also connected with a documentation local and cloud hosted documentation strategy so active inference institute dot github dot io slash start will give

you some hosted documents and also they will pop up locally and some quick links are shown to run the experience you can do dot slash run dot sh that's what we did or let's open up another window and use the other top level run script which is dot slash run underscore docs dot slash run underscore docs so what that's going to do is run the document checking and rendering preparing any packages that are needed along the way and then pop up here in the browser window is a locally rendered at ip address 127.0.0.1 port 8003 so this is the locally implemented

server for documents or you can look at the exact same material in the github document pages server for documents or you can look at the exact same material in the browser

and we'll kill the dock runner

okay

just crashed

okay

okay we're back

okay

we're back

all right so

so

we're live in the stream i look forward to anyone's comments and let's just go over what is going to happen first what is the start here repo well this is just to get it out in an inference stream and show a little bit of the behind the scenes but this repo broadly speaking gives an interactive and well-linked landing page into the active inference ecosystem and institute now elsewhere we have some welcome page that has a lot more interactive material and a lot of hyperlinks to our our living document but there isn't any code execution in coda so what i've been working on is this start repo which has a little bit more of a seamless experience in terms of not just linking to resources but actually making a repo that runs them so this repo gives a fun interactive matrix themed uh landing page into the the institute and ecosystem it renders as we've seen into local documentation as well as updated github pages

it uses generative ai to write custom curriculum we'll see if that worked that looks like it's still going and it also helps us clone and deploy repositories so for example what we have cloned while that was all happening is active inference dot jl which i have made a fork and some modifications to axiom which is the recent open source architecture by versus cerebrum which is the linguistic approach to modeling case systems and other linguistic aspects in models the cognitive repo which is a large underlying obsidian formatted markdown knowledge repository that you can open up with foam or in obsidian generalized notation notation gnn for notation and metaprogramming with generative models lean niche which has been a bit of a fun collaboration with jason larkin and we've been using the lean theorem prover to explore a few aspects of probabilistic dynamical systems in the fep pymdp which i've made some heavy modifications to that we'll go through rx infer examples also a fork of which i've heavily modified and template repo which is a repo for setting up a scientific paper rendering

pipeline involving code and figures and equations and all that so this is just an example you can add more repos that get cloned in but start is hosting all of these different repositories as options to download in so those repos just mentioned and many other implementations exist can help bring together open source implementation type frameworks like pymdp active.jl rx infer.jl axiom etc and all kinds of emerging tools be they related to synthetic intelligence llm etc and also anything that you want to bring in in terms of adding in another custom repo or just having another repo in this clones folder attack of the clones so that is a cool setup landing page nexus there's a few themes and ideas that i think we'll revisit it but what i will do next is look at these themes get a first pass on the themes then we will look at some of this custom

generated curriculum material then we will turn to the pymdp and active.jl forks where i have made the most of the modifications

so themes and ideas start here now or act and first serve whatever you prefer just jumping in active is about action

and perception it's about inbound outbound about joint distributions for agents and niche grip or gripped duck rabbit all this kind of stuff that we're always talking about and in our own learning and application and and contribution journeys we just start here now or there now or then if that's a better one but it is about doing it is about starting where and how you are not waiting for something else to be the case unless that's where and how you are cartographic pluralism the name of the stream is a map not the map it is a map it's not the only map it's gonna version even during the stream and also cartographic realism cognitive realism open source intelligence ecosystems and we're talking about ecosystems that are we're mapping the ecosystem in terms of sharing links and resources and signposts so we're mapping the ecosystem

of maps of maps of minds like mice people ant colonies llm that map minds or not might might map another mind

there or it could just map something we're mapping the ecosystem of maps of minds generative models that do

mapping there are multiple implementation frameworks for active inference there are many ways to build up along the low road we see that just by the juxtaposition of all these implementation frameworks right here and with gnn we'll even have each of them implementing where possible some of the exact same models and state spaces so even for a given state space there might be one or more accessible ways to implement it with different trade-offs so implementations do not negate each other in active inference shipping the docks and docking the shipment whether we approach this from a vibe coding or kind of a developer perspective or just the relationship between making sure that we're documented all along the way and documenting what we ship and shipping the docks and this idea of notes to navigation and the ways to include

the things that we're noting along the way in our field work and get through some really dense sites in one sitting okay clone the repos let's close that window

all right let's exit the matrix

so the generated content there is in the learning

data so the domain research

data

here's the coffee tasting research that was using perplexity

i will push this to github just so we can look at this a little bit

in a rendered form

data

data

here was in france stream 8.1 that was from december 15th 2024 and that was uh

oh it was uh 8.1 symbols greetings and that was the first iteration of the start repo

okay so here's the audience research here's tulsi gabbard we also have that domain research with the coffee tasting research

and then here's the written curriculum

here's the tulsi gabbard curriculum here's the complete complete tulsi curriculum

so it includes a little bit of information on the person's background

their motivation what kinds of core materials we're going to share with them

what kinds of assessments how to connect it to the person's background

so this is just a starting point for of course that person or for any person or for the system that generates it but it gives some really fun and exciting starting points

so let's see a little william blake

background again it's to be discussed explored played with

how much do you want to meet william blake where he's at versus just have a sort of

um attractor that isn't personalized so it's not that there's a

a wow look at this look at this gibberish part

incredible um it and uh then we have the translations

so that is in translated curriculum it looks like the the translation stage is pretty slow i didn't run it by default but we can see that there's these translations of coffee roasting and so on okay

here we go starting to look a little bit at the uh live chats by way of oh it's still writing the uh still writing some of the material okay so

let's go to some of the live chats

okay so dean hello dean wrote i have no technical background with any of these repos but i do want to take advantage of the curriculum generator what's my first step do i need to learn how priors with active no you don't need to have priors with active you need to have it just this is this starting point

clone this repo start find out from help or ai or videos how to use github clone the repo go to the the environment example file copy it into a file called dot env that just then add in in the real env

which is your secret file add in the api keys for at least perplexity in open router then top level of the repo dot slash run dot sh

it's going to go through these setup phase these this is not a loading screen i think we can put we can put different custom messages there

here

i wouldn't say it's overly dramatic okay then when it pops up here we are in that interactive terminal

three generate curricula

so we'll just get this uh chugging along because it's it'll take a little bit in the background while

we address the other one of your comments dean and look at some of the repos themselves okay select domain

let's do

islands

navigation using boats

yes

new domain target entity this is it could be skipped or you can come up with another entity um

i'll do bill gates

uh

employee at microsoft and

!

i'll just call them a common character

target language and here we'll do

ancient latin

yes create that new one

proceed

all right

so that's all that takes

and a little bit of time

to generate a new curriculum

and this is just the version that we have today so future versions you can imagine a

graphical interface or

um

um

we can even make that right now

so here add another script

in this folder

calls

generate

curriculum

gui

which pops up a simple

browser window

with
the same fields and information
as the interactive terminal one
yielding
a
gui for
custom active inference
curriculum
generation
and then
while it is
working
it says
working
with status bar
estimate
and then
when final results
are ready
it
shows them
in
gui
thanks gpt5
all right
so we have
the bill gates curriculum
developing in the background
and meanwhile
i'm going to quickly review
a few of these forks
of cloned repos that i've been working on over the last few days
and we'll keep an eye out on anyone's comments questions in live chat
all right so here we go
go
the uh
well there's several interesting ones they're all very interesting
but i want to show mainly the pymdp so this is all within this pymdp fork

!

and just to show where this is in the browser

this is

the

doxology

so my personal github account fork of pymdp

!

and the branch

!

is called textbook

so all that information is open source available

okay

but here we'll look at it through the

interface

of start

opened up in cursor

and we are in cursor 1.59

okay

so a couple of modifications and things in this textbook

uh

first is the examples from pymdp

which is just a

copy over from the examples

like the agent demo and grid world

copy them over in their most minimal form

and then the examples from pymdp

underscore modified

which is where

uh

modified versions of the scripts

are made

that output more data

and more visualizations

than

the

uh

notebook forms

do

so

if you want to

the output of all of the

scripts are run there

so

purely just for getting a better handle on the pymdp examples

and to understand

what they

look like

that examples from pymdp modified

is pretty cool

pretty useful

very open to further developments that people can do

in of course the pymdp examples in the underlying repo or their own forks

or improving and making these examples more flexible

but the

examples

so just examples

is where i have made 12

new examples

they get run by this script called run all

so run all.sh

will run these 12 scripts

and here are the outputs

so this is about to be like a

rapid

review

kind of like the 2022

part all textbook

that brings together

the fundamentals

of

probability

and the base rule

in module 1 and 2

then

module 3

4

5

and 6

are related to

sense making and perception

module 7

8

and 9

are about active

inference

policy inference

and then

examples 10

11

12

are relating to

the

the

p

o mdp

the partially observable

markov decision process

so here's what the outputs look like

starting from

one

probability basics

this is going to talk about

a few different distribution

types

so here's a categorical distribution

on the top

three

discretely different options

the gaussian or normal distribution

in the center row

and then a poisson

or a rate parameterized distribution

on the bottom

and each of those are shown in
three different forms
in terms of three different
parameterizations
so it's like
categorical
you're talking about things that
differ discretely
and all of them across the
categories sum to one
and that could be like
a third a third a third
or it could be
bimodal
or it could be peaked
gaussian
you can move around
on the x-axis
or you can make it
narrower or wider
that's variance
or precision
and then this is
a few of the different
lambda parameterizations
the poisson
in module two
bay's rule
there is
this is a summary
of the information
gain and variational free energy
for two different settings
the first setting
is the classic
medical diagnosis
setting
where there's a prior belief

about how likely
a disease is to be present
which could be
the population
overall frequency
it could be some other
prior belief
and then there's a test
which has a 90%
sensitivity
true positive
and a 95%
specificity
true negative
and so
when you get a positive test
conditioned upon
the observation
of a positive test
not to overly simplify
but the frequentist
says I mean
90 or 95%
that the person
has the
disease
because that's what
the accuracy of the test is
so that's what the most likely
outcome is
whereas here
with the base
we can see
well depending on
your prior
that's going to update
what happens
when you

update after the
posterior
so that's
shown here
and then in the
sequential example
is weather
where you don't know
whether
it is raining or not
I guess you're
you're trapped inside
of a
of a bunker
or something
and you're just seeing
whether
umbrellas
or no umbrellas
are going with
people walking by
and so you have an initial prior
70% likely to be sunny
and then you see
sequences of umbrellas
or not
so that's
kind of
background
statistics
good info to know
module 3
and
the sort of
cousin of 3
refactored 3
these have to do
with observation models

in PiMDP
so
these are
ways
sometimes called
the A matrix
or the ambiguity matrix
where you have
the mapping between
observations
and latent states
and that can be
like the identity matrix
which is to say
a noiseless
perfect mapping
so basically
just like a
Markov decision process
not a
partially observable
MDP
or
different levels of
noise
off diagonal
symmetric matrices
all these different things
state inference
uses the
infer.states
method of
PiMDP
to
use variational
methods
to update
priors into

posteriors
given observations
sequential inference
in the fifth module
is a little bit more
about different patterns
of sequential evidence
accumulation
which
is worth
going into
so
for example
initially
we have
a uniform
prior over
latent state
A B or C
one third each
first time step
we see observation
zero
this updates
our belief
that it's
80
15
0.5
second observation
time step
comes in
it's zero
again
we increase
our
posterior
belief

from 0.8
to 0.96
then
third observation
coming in
observation
one
so then
that goes
from 0.03
to 0.22
and for each of
these
we can calculate
the information
gain
per update
we can look
at the
components
of the
variational
free energy
we can look
at the
validation
of the
normalization
of pi mdp
that's five
six is a
beginning point
for a multi-factor
example
so here
there's two
independent factors
there's the

weather
and the
location
so three
possible locations
two factor
two um
options for the
weather
factor
and then
that sets up
six possibilities
and so there's
different beliefs
over joint states
okay
so that's getting
through the
sense making
stuff
seven eight
nine are
about action
inference
so seven
gets you a little
bit familiar
with the b matrix
or the transition
matrix
in terms of
it being
modeling from
to
latent states
but no
action

selection
yet
eight
is going to go
into action
selection
using preferences
and pragmatic
value in the
expected free
energy
so this
setting
there are three
different options
for location
left center
and right
the c or the
preference variable
is set
so that the
left has a
negative preference
it's disfavorable
the center
is neutral
and the right
has a positive
preference value
and then we can see
how conditioned
upon your starting
state
different moves
have different
pragmatic value
here the epistemic value

zero for all
and then we can see
how softmax
over the negative
expected free energy
yields updated
action
posterior probabilities
that get sampled from
from the same
action
prior
which is the habit
or the e matrix
so for example
if you're starting
in the left
state
then move left
which in this situation
keeps you in the left
or stay
they lead to the same
expected outcome
observation
so they are equally
likely to be selected
whereas right
has a
higher
preference
zero higher than
negative two
so that
has a higher
it's more expected
to be the case
this is

how
we
get
preference
and
goal
oriented
behavior
without
a reward
distribution
or
reinforcement
learning
to training
data
is
we have
things
do
what
they expect
they
they
change their
mind
and change
the world
to observe
what they
expect
here
starting in
the center
it's
especially
disfavored
to go to

the left
somewhat
disfavored
to stay
highly favored
to go right
and then
for starting
in the right
position
we see
that there's
again
the same
normalized
probability
of selecting
either
moving right
which is
staying
or staying
which is
analogous to
the starting
on the left
situation
and staying
left
so
that is
the eighth
module
which gets
in
and also
this is a
separate one

in eight
talking about
like
different preference
distributions
for the sweet
lover
and the
health
conscious
taster
tm1tap
wrote
wtf is
going on
here
and why
is it
recommended
to me
yeah
let me
know
when you
find out
okay
ninth
module
is a
similar
example
where
uh
i don't think
the coloration
on the heat
maps is all
perfect

the epistemic
status of all
of this
is
functional
and
improvable
so
i'm sure
you
and
our
synthetic
intelligent
viewers
will find
many places
to improve
please make
that make
that contribution
make that
github
request
so here
we're in a
situation
where there's
four
states
and the
left three
are equally
neutrally
preferable
and the
right state
is

preferable
so then
we can see
that just
from reactive
planning
we
rarely end
up with
a reward
but if
we're able
to plan
and look
ahead
we can
it's a
trivial
example
but it's
just
demonstrating
how the
expected
free energy
is used
for pragmatic
value
and then
10, 11, 12
go through
the
POMDP
example
and shows
the matrices
and
how that

state space
gets set up
and then how
beliefs evolve
through time
what action
and observation
sequences
happened
10 is
the
T-Maze
setting
11 is
the
grid world
some blank
visualizations
there
grid world
setting
and
12
is
another
T-Maze
with a bit
more
and different
information
so
that is
intended
to be
useful
resources
again
that's

the
doxology
fork
of the
PYMDP
repo
textbook
folder
examples
are the
12 examples
that I
wrote up
with help
obviously
and then the
examples from
PYMDP
modified
if you want
to stick
closer to
the
general purpose
examples
in
PYMDP
then those
are the
ones to
go with
okay
let's see
how the
okay
let's go
over to
the

updates
in
active
inference
dot JL
so
active
dot JL
here
this is
a
Julia
based
package
and
I have
also
made
a
textbook
branch
and
sub
folder
so
in
this
case
there
are
five
examples
so
it's
structured
a little
bit
differently

the
first
example
has to
do
with
the
mathematical
foundations
second
example
this is
just
another
just
a
slightly
different
learning
path
second
module
has to
do
with
the
variational
free
energy
third
example
expected
free
energy
fourth
example
is
going

to
use
the
PYMDP
fifth
example
hierarchical
inference
and
multi-agent
coordination
and
here's
what
the
outputs
look
like
so
first
the
foundations
that
curriculum
generation
is still
happening
in the
background
it's
pretty
slow
okay
so
here's
the
from
the

foundations

module

in the

active

jl

repo

fork

pretty

old

computer

I'm

running

it

from

pop

os

maybe

it'll

be

better

to

let's

look at

it

within

the

browser

textbook

!

output

okay

so

there's

the

information

theory

so

here

we
have
four
different
distributions
observation
comes in
and
then
we
have
KL
divergences
calculated
I'll
just
do
it
here
VFE
the
seconds
unit
yeah
little
slow
but
c'est
la vie
like
it's
faster
to
load
it
through
github
than

to
load
it
in
the
IDE
tragic
here
is
the
VFE
perceptual
inference
state
inference
process
we'll
just
let
that
one
load
okay
similar
examples
in the
EFE
module
we have
analogous
examples
for the
EFE
and
same
as the
pomdp
and the

multi-agent
so
kind of
fun
and then
we'll
look at
the
multi-agent
trajectory
okay
we'll get
this
here we go
so this is a
multi-agent
grid world
metagrid
is probably
a better
framework
for doing

this kind of
collective
foraging
multi-agent
situation
I'll just
bring that up
in the
browser
yeah
David
blumen
et al
metagrid
is

is
better
if you
actually
want to
make like
these
kind of
role-playing
game
meets
collective
foraging
multi-agent
flexible
active
reinforcement
learning
settings
okay
here we
go
now the
GUI
has popped
up
so this
was again
prompted
just
just
that was a
two-shot
prompt
first just
said
make a GUI
for what

does the
same thing
as the
typing
so let's
domain
nursing
entity
Elon Musk
language
Russian
so there
we go
all right
just to
close off
there
if you
really want
to do a
multi-agent
setting
and you
want something
that's really
professional
metagrid
or some
other option
is probably
a better
choice
but the
whole point
of what's
happening
here
is we

can look
at the
expected free
energy values
and posterior
beliefs and
all this kind
of stuff
and we
can connect
with how
we learn
and apply
these
implementation
frameworks
okay
so that
works
so that
works
all right
now I'm
going to
return to
Dean's
previous
questions
and his
new one
so yeah
if anyone
else has
any
ideas
or
questions
or they

want to
see something
happen
okay
whose map
am I
building
whose map
am I
building
the curriculum
that modifies
a current
mind
model
so map
outside both
the curriculum
designer and
the curriculum
subject
is the
limit of
my self
organization
my ability
to generate
effective
prompts
it seems
like I'm
giving most
of the
organization
tasks over
to the
processing
agent

which isn't
me
true
and then
he wrote
I see
the AI
generating
and organizing
shouldn't
I practice
these tasks
too
yeah these
are these
are huge
questions
are definitely
things that
I've been
thinking about
over these
last few
days
working on
it
let me
have a
sip of
water
and I'll
think about
what
tack to
enter
here
shouldn't
you practice

those tasks
too
well
to the
extent
that we're
going to be
normative
here
pedagogically
yeah I
think that
you should
one should
I should
etc
generate
and
organize
and
while the
AI is
doing a
lot of
the
terminal
generating
and organizing
in terms of
writing the
actual curriculum
itself
first off
those are
just intended
to be
intermediate
artifacts

produced
for curation
and secondary
refinement
so it's
not that
it's generating
the final
word
it's
generating
a blob
of clay
that might
be
close
or far
from
your
vision
and it
is organizing
things in a way
that has to be
quite strongly
typed
in order to
find exactly
like to use
the entity
research
on
some data
to use the
the entity
research
on Barry
Bonds

yes it has to be
organized into the
right fields
and all of that
but there's a
higher level
organization
that is
organizing
that
kind of
syntactic
organization
and then there's
the syntax
and the semantics
of generating
these systems
that do
this thing
and that is
its own kind
of spring
training
practice
is the limit
of my
self-organization
my ability
to generate
effective prompts
yeah there's
there's a lot
of layers
to look at
that question
at
in some

ways
you can ask
for an
effective prompt
on
some
you can ask
for a kind
of prompt
like write a
good prompt
for doing this
and that's like
off-sourcing
more of your
exact semantics
in exchange
for
what might
yield
more
compliance
output
especially
structurally
but
only you
are going
to be able
to generate
what you
can do
however
aren't we
seeing that
with even
a
few words

or an
image
it's
possible
to
mimic
quite well
it seems
like I'm
giving most
of the
organization
tasks over
to the
processing
agent
which isn't
me
yeah I
think that
well
first I
wonder
to what
extent
was that
thought
in the
calculator
days
or in the
steam engine
days
or in the
early
automobile
days
or the

video
recorder
days
or the
excel
early
days
or
whatever
it is
we are
doing
different
things
okay
is limited
by self
organization
my ability
to generate
effective
prompts
short answer
no
prompt

skill helps
with the
ceiling
on self
organization
is set
more by
your
systems
than by
phrasing

alone
how do
you raise
the ceiling
define
artifacts
plus
DOD
probably
not that
one
but
interesting
add
tight
loops
use
scaffolds
externalize
context
that's
kind of
the
context
engineering
angle
stable
interfaces
automate
verification
prompts
are one
lever in
that
system
useful
but not
the

limiting
factor
huh
Bill Gates'
SAT score
um
yes
the GUI's
still plugging
away
metagrid
for multi
agents
translations
into
custom
languages
local
and
github
documents
for a lot
of machine
readability
and we can
translate these
documents into
other languages
as well
uh
okay then
just briefly
the last of the
implementation
framework
where I've been
adding a lot
of work

is in the
src
clones
oh it
didn't
it didn't
clone
rxinfer
I'll just
show it
uh
in the
browser
here in
the
research
folder
on my
branch
or my
fork
of rxinfer
examples
are
several
fun
uh
methods
related to
rxinfer
uh
generalized
coordinates
hierarchical
gaussian
filter
integration
with

Ilm
upon
dp
the
infinite
data stream
situation
visualization
methods
and then here's one
that I will show
which is the
manim
package
so
uh
people may be
familiar with
the
three blue
one brown
math education
channel
who makes
videos like
this
and there's a
open source
package
called
manim
and so
I use the
manim
package
to
go through
all of the

rxinfer
examples
so it's like
we have this
kind of like
mini documentary
procedurally
generated
starting point
for each
of the
models
in rxinfer
so there's two
different styles
of model
uh
one of them
is gonna have
a
uh
I'm just picking
two random
examples
there's the
uh
drone
uh
there's the
uh
source code
that's just
gonna have this
like
source code
kind of pop up
on screen
and then there's

the
dot graph
so it's
not gonna
display in the
but
it's a good
starting point
uh
so
extra
usable
setup
utilities
and examples
in
activejl
rxinfer
dot jl
and in
pymdp
uh
recapitulating
the examples
in those repos
and adding
these textbook
folders
and each
of those
are just
cloned
sub repos
within
this
start
repo
the

start
repo
has the
ability to
generate
more
materials
like we're
seeing
and then
also it
has this
start
here
dot md
so the
idea is
welcome
this is a
simple
actionable
landing page
into the
active inference
ecosystem
and institute
centered on
this repository
if you only
read one
file to
begin
make it
this one
why this
repo
exists
learning by

doing
generate
visualize
and translate
curriculum
content
grounded in
real data
build end
to end
no mocks
real io
reproducible
scripts
and tests
and contribute
to the
ecosystem
improve docs
!
scripts
and learning
materials
for the
broader
community
here's a
bunch of
links
the obsidian
link
is the
one
is the
live sync
version
of the
cognitive

repo
so if
you want
to
play
around
in this
kind
of
knowledge
graph
that's
the
cognitive
repo
you can
just
clone
it
in
open
it
locally
this
here.md
talks about
a few
things you
can do
with the
repo
and then
talks about
how you
set up
the repo
with the
uv

package
manager
and run
the
run
docs
and
the
run
also
just a
starting
point
but
gets a
lot of
it
there
and
you
can
always
go to
welcome
or obsidian
for that
kind of
multimedia
linked
living document
style
however I
think that
the start
repo
is
filling a
critical

service
and role
in the
open source
ecosystem
which is
this kind
of
meta
repository
management
having
setup
scripts
that
install
uv
if you
don't
have
it
then
use
uv
to
set
up
the
environment
and then
use
that
set
up
environment
to
clone
repos

forks
of those
repos
which
themselves
have
setup
scripts
and
so
here's
the
setup
.sh
within
the
pymdp
fork
so
by
running
this
sequence
of
setup
scripts
you
will
go
from
either
a
click
or
just
one
line
entered

into
this
cloned
repo
all
the
way
on
through
having
checked
files
that
run
and
do
different
aspects
that are
important
for
understanding
conceptually
and
building
an
active
inference
so
here's
the
axiom
repo
by
verses
and
then
we

can
just
go
right
in
there
and
just
say
okay
run
the
axiom
repo
right
from
here
I
have
a
private
fork
of
axiom
as
well
but
this
is
just
let's
just
see
what
happens
with
one
shotting

it
here
okay
so
talked
about
the
repo
about
how
it
can
generate
material
talked
about
a few
of
these
implementation
strategies
covered
up
for
these
three
pymdp
active
jl
rx
infer
jl
about
the
forks
that
I
have

been
working
on
that
are
facilitating
setup
and
examples
for
a
later
date
maybe
with
the
GNN
lean
niche
cerebrum
and
I
think
it
takes
us
right
up
to
the
edge
so
just
for
that
one
line
now

we
are
running
axiom
so
it's
playing
that
video
game
explode
I
don't
know
what
that
is
but
it's
a
video
game
and
that's
the
beauty
of
when
a
repository
has
its
own
setup
materials
and
reproducibility
now we

can just
clone
axiom
and run
it
so
like
so
many
cool
ways
to
compose
across
repos
cross
render
in and
out of
repos
public
repos
private
repos
there's
rx
and
for
examples
with
neural
networks
a lot
of
work
across
different
repos

in
llms
and
other
other
ai
methods
so
for
those
who
are
curious
or
soon
to be
curious
about
how
do
you
learn
and
apply
synthetic
intelligence
and
active
inference
that's
why
we
have
a
map
start
slash
here

and
then
feedback
along
the
way
just
what
parts
were
challenging
which
parts
were
awesome
what
you
like
where
did
you
get
held
up
did
you
like
being
held
up
where
would
you
have
wanted
something
different
and

then
either
raise
your
hand
and
give
the
feedback
or
reduce
your
uncertainty
by
making
that
change
yourself
and
figure
out
how
to
fix
it
on
your
own
local
version
and
you'll
be
contributing
to
open
source
software

in
no
time
okay
let's
see
how
that
educational
material
went
while
Axiom
is
running
we'll
do
it
through
github
okay
just
push
it
all
right
thank
you
dean
oh
it's
in
data
it
used
to
be
in

the
learning
folder
okay
audience
research
translate a
curriculum
visual
up
we'll
just
look at
it
in
the
browser
translated
okay
here's
from one
minute ago
ancient
Greek
so here's
the
myrmecology
curriculum
for active
inference
translated
into
ancient
Greek
I will
leave it
to
I will

leave it
to
a speaker
or reader
to
evaluate
here's the
audience
research
as a
JSON
and then
here's the
written
curriculum
for those
domains
and entities
so let's
see
we have
the
Elon
Mosque
and
the
island
navigation
using
boat
then I'll
just see
if there's
any last
comments
otherwise
that'll
be it

okay
again
we can
update the
prompts
we can
have
mermaid
diagrams
we can
have
more
links
we can
base
it off
of
whatever
material
is put
into
synthetic
FEP
actinth
this is
the material
that's
used
to
generate
so
that
can be
versioned
and
yes
careers
and

educational
foundations
in
island
navigation
wow
perplexity
or open
rodder
I don't
know which
one of
them is
yielding it
probably the
open
rodder
look at
some of
these
I think
it's using
the
like the
llama
scout
for
or something
like that
but
whoa
as recorded
class expenses
valued
everyone
machine
grand
blind

likewise
quotes
request
conscious
integrity
understand
entered
podcast
associate
straw
credential
troll
kept
hypotheses
streamline
incidents
near
longitude
acquainted
dissipate
approach
cold
soil
preventing
utilize
sense
modification
deterview
replication
planner
legal
cool
all right
I'll just
wait one
minute unless
anyone has a
final comment

or question
as usual
I hope
this is
interesting
or
fun
for you
I
put all
of these
out
in the
open
source
spirit
for
people's
feedback
their
commenting
their
engagement
there's
a lot
of
fractal
ways
to
contribute
you
could
make
your
own
repo
and
then

we
can
have
it
cloned
or
you
could
make
a
pull
request
or
raise
an
issue
right
here
in
the
repo
so
that is
all
for
tonight
I will
leave
to
you
to do
the
who's
on
first
and
all
of

the
dramatic
readings
and
everything
just
wanted
to
get
that
out
there
hope
that
this
is
a
place
where
you
can
say
where
should
I
go
where
do
I
return
and
we
can
make
these
links
accessible
rigorous

applicable

so

thanks

for

watching

along

and

enjoy

with

be

will